Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 1b - Modeling the Environment Dynamics

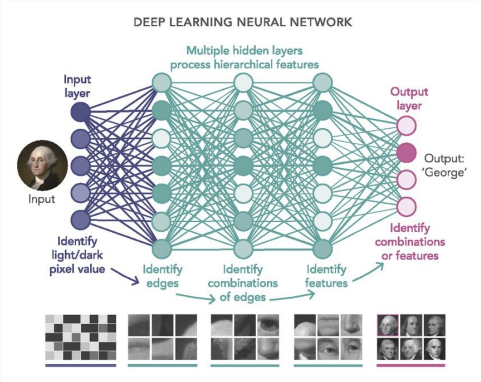
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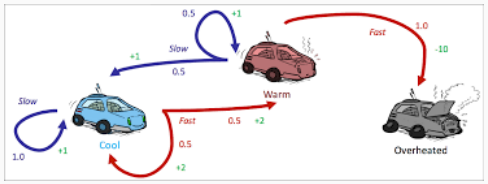
Modeling the Dynamics of the Environment

Terminology

- In ML:
  - **Model** is used to represent the structure of the network used for classification / regression.
  - Part of the output of the algorithm



- Here:
  - **Model** denotes the representation / abstraction of the environment.
  - Part of the input



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Abstractions

- **Motivation:** Real-world decision-making problems are often too large, too detailed, or too uncertain to reason about directly.
- **Abstraction:** A mapping from a detailed (low-level) model to a simplified (high-level) one that preserves information relevant for decision-making.
- **'Penalty' of abstraction:** Loss of information $\Rightarrow$ potential suboptimality

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Abstractions





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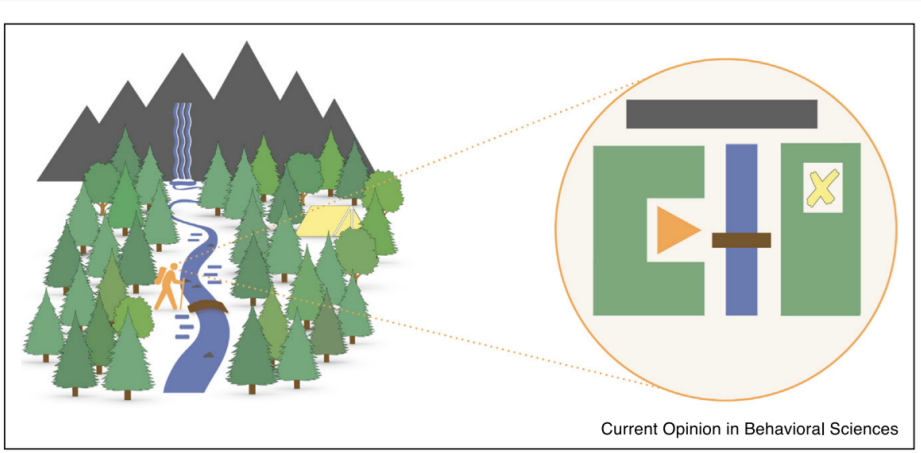
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Abstractions





Images from
https://david-abel.github.io/index.html

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Types of Abstractions

- **State abstraction:** Grouping concrete states into abstract states
  - Example: grid cells $\rightarrow$ rooms
- **Action abstraction (temporal abstraction):** High-level actions that encapsulate low-level control
  - Example: “navigate to door” vs. primitive moves
- **Model abstraction:** Simplifying dynamics or uncertainty
  - Deterministic model instead of stochastic
- **Objective abstraction:** Reasoning over abstract goals or tasks instead of raw rewards

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Types of Abstractions

We will use ϕ to denote abstractions, and ϕ^{-1} to represent the

For example, state abstraction $\phi : \mathcal{S} \mapsto \mathcal{S}'$ such that $\phi(s) \in \mathcal{S}'$ is the abstract state corresponding to ground state $s \in \mathcal{S}$, and the inverse image $\phi^{-1}(s')$, with $s' \in \mathcal{S}'$, is the set of ground states that correspond to s' under the abstraction function ϕ.

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Environment Model: States

State Space: $s \in \mathcal{S}$ is a world state (more accurately, a representation of a world state)

Typically, it is impossible / impractical to explicitly specify the full state space.

We therefore use an abstraction we refer to as a **factored representation** in which the state space is described via a set of variables $\mathcal{X} = X_1, \ldots, X_n$.

A state is then an assignment of a value $X_i \in Dom(X_i)$ for each variable X_i.

Why is this an abstraction? Is it a 1:1 mapping?

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Environment Model: Dynamics

Environment dynamics describes how the state of the world evolves over time.

Dynamics specify the relationship between consecutive states, s_{t+1} given s_t

We distinguish between:

- Deterministic dynamics
- Stochastic dynamics

These dynamics are properties of the environment, independent of any agent behavior (which we will account for later on).

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Deterministic Dynamics

Deterministic Transitions

- The next state is uniquely determined by the current state:

$$s_{t+1} = f(s_t)$$

- No uncertainty is modeled in state evolution
- The same initial state always produces the same trajectory

$$\tau = s_0, s_1 = f(s_0), s_2 = f(s_1) \dots$$

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Example - Newtonian Dynamics

Newtonian dynamics with known forces.

Determinism comes from: exact forces, exact parameters, and no noise

Newton's laws:

$$\frac{dx(t)}{dt} = \dot{x}(t), \quad \frac{d\dot{x}(t)}{dt} = \ddot{x}_t$$

Where $t \in \mathbb{R}_{\geq 0}$ is continuous time, $x(t) \in \mathbb{R}^n$: is the position, $\dot{x}(t) \in \mathbb{R}^n$ is the velocity, and $\ddot{x}(\cdot)$ is the (deterministic and known) acceleration law.

These differential equations define a continuous-time dynamical system.

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Discretization

Physical systems are often described in continuous time, but reasoning and computation are not.

We discretize continuous dynamics in order to:

- Enable algorithmic reasoning and simulation
- Obtain a finite or countable state-transition model
- Match discrete-time sensing, actuation, and computation
- Apply planning and learning algorithms

Discretization is an **abstraction** that trades accuracy for tractability.

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Discretization over Time

- Continuous-time models require solving differential equations, but closed-form solutions are often not available
- Discrete-time models are obtained by **discretization**:

  $$x_{t+1} \approx x_t + \Delta t \cdot \dot{x}_t, \quad \dot{x}_{t+1} \approx \dot{x}_t + \Delta t \cdot \ddot{x}_t$$

  This is a **factored representations** where:
  - x_t: is the position at time step t,
  - $\dot{x}_t$ is the velocity at time step t,
  - $\ddot{x}_t$ is the acceleration, determined by physical laws (e.g., gravity, forces)
  - and Δt is the discretized time step
- The choice of Δt controls the trade-off:
  - Smaller $\Delta t \Rightarrow$ higher accuracy, higher cost
  - Larger $\Delta t \Rightarrow$ lower cost, more approximation

Discretization is an **abstraction** that trades accuracy for tractability.

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From Deterministic to Stochastic Dynamics

- Deterministic models abstract away uncertainty.
- In practice, forces are approximate, parameters are unknown, and unmodeled effects are present.
- As a result, the same state may lead to different future states.
- We next move to **stochastic dynamics** that explicitly model this uncertainty using probability distributions.

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Probability 101

Notation Conventions

- **Random variables:** uppercase (X_t)
- **Realizations/values:** lowercase (x_t)
- A random variable takes values according to specific probabilistic laws
  - In robotics: sensor measurements, controls, states
- **Probabilistic inference:** the process of calculating these laws for random variables based on other random variables and observations.

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Notation Conventions

$$\Pr(X = x), \quad \Pr(X_{t+1} = x' \mid X_t = x),$$
$$\Pr(X' \mid X), \quad \Pr(A \cap B)$$

- Acts on *statements* (events)
- Refers to the probability measure
- Used for conceptual reasoning about uncertainty

$$p(x), \quad p(x' \mid x), \quad p_\theta(x)$$

- Function or density
- Can be parameterized, learned, integrated
- Used for modeling and computation

Discrete: $\Pr(X = x) = p(x)$ Continuous: $\Pr(X \in A) = \int_A p(x)\, dx$

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Basics Concepts in Probability

Discrete probabilities sum to 1

$$\Sigma_x \Pr(X = x) = 1$$

All probabilities are non-negative

$$\Pr(X = x) \leq 0$$

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Basics Concepts in Probability

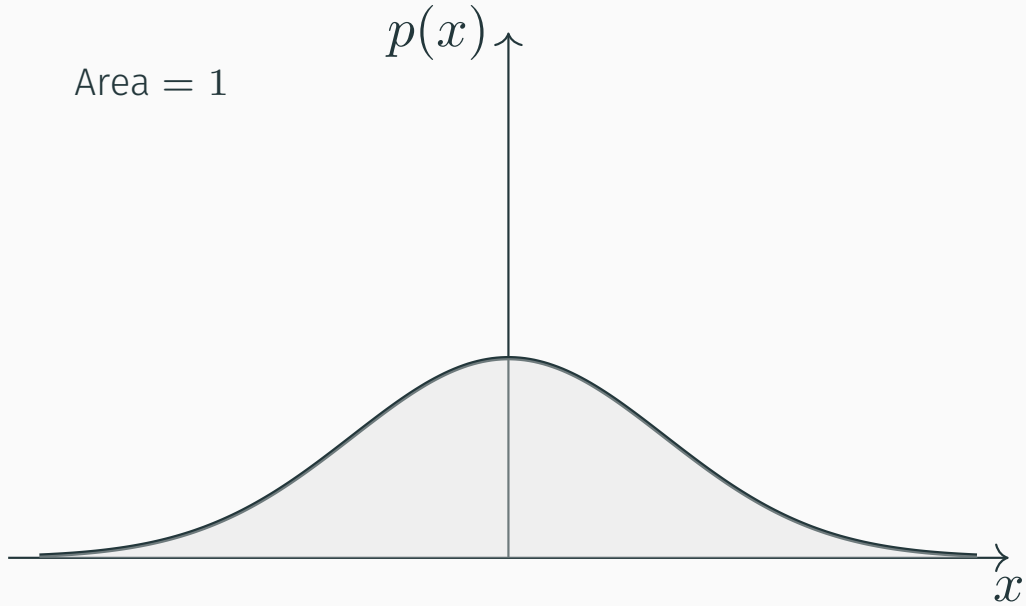
Probabilities in **continuous space**:

- Random variables take on a continuum of values according to **probability density function (PDF)**
- $p(x)$ measures *density per unit of* x, not probability, so it can take values that are greater than 1.
- The **area under the curve** must equal 1 (not the height of the curve).
- Rule $\Pr(a \leq X \leq b) = \int_a^b p(x)\, dx$
- A PDF always integrates to 1

$$\int_{-\infty}^{\infty} p(x)\, dx = 1$$

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Basics Concepts in Probability



A common PDF is the normal distribution.

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Normal (Gaussian) Distribution

Definition:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

- μ: mean (expected value)
- σ^2: variance (spread)
- Symmetric bell-shaped curve centered at μ
- $E[X] = \mu, \quad \mathrm{Var}(X) = \sigma^2$
- **Multi-variant:** Normal distributions over vectors:

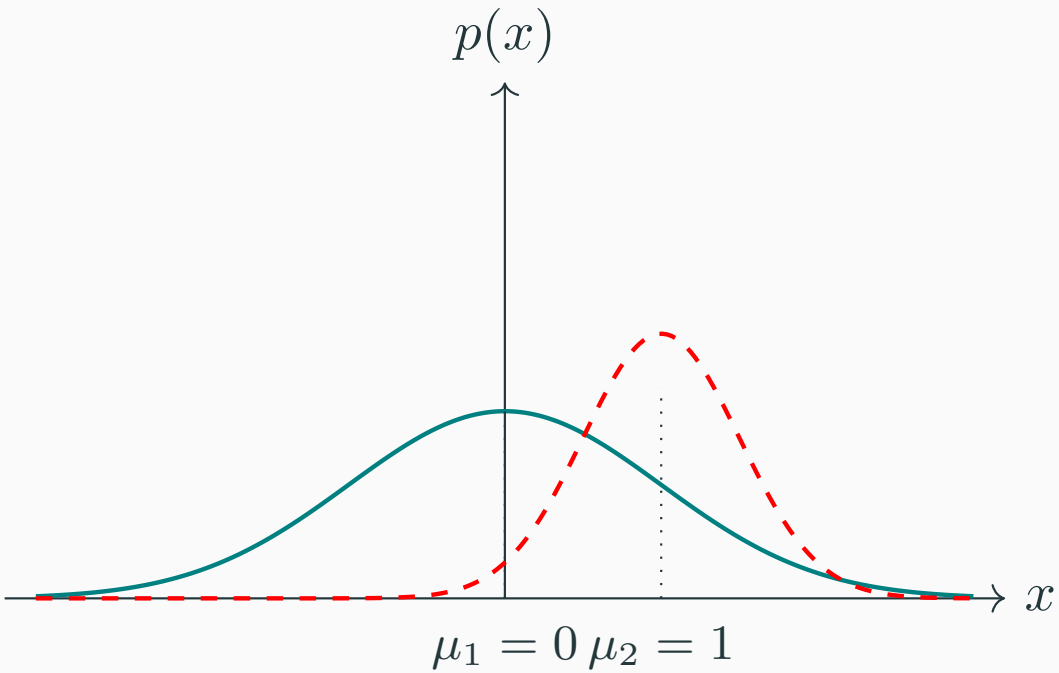
$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left[-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right]$$

Here, μ is the mean vector and Σ is the **covariance matrix**.

$$\int_{\mathbb{R}^n} p(\mathbf{x})\, d\mathbf{x} = 1$$

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Normal (Gaussian) Distribution



Solid: $\mu = 0, \sigma = 1$ Dashed: $\mu = 1, \sigma = 0.7$

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Joint and Conditional Distributions

Joint Distribution

$$p(x, y) = \Pr(X = x, Y = y), \qquad \sum_x \sum_y p(x, y) = 1$$

Chain Rule:

$$p(x, y) = p(x \mid y)\, p(y)$$

Conditional Probability

$$p(x \mid y) = \frac{p(x, y)}{p(y)}, \qquad p(y) > 0$$

Independence

$$X \perp Y \iff p(x, y) = p(x)\, p(y)$$

Conditional Independence:

$$x \perp y \mid z \iff p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$$

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Marginal Distributions and Independence

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Marginal Distribution

- Discrete:

$$p(x) = \sum_y p(x, y), \qquad p(y) = \sum_x p(x, y)$$

- Continuous:

$$p(x) = \int p(x, y)\, dy, \qquad p(y) = \int p(x, y)\, dx$$

Theorem of Total Probability

Discrete form:

$$p(x) = \sum_y p(x \mid y)\, p(y)$$

Continuous form:

$$p(x) = \int p(x \mid y)\, p(y)\, dy$$

This expresses the marginal distribution of X as a weighted average of its conditional distributions given Y.

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Bayes Rule

Discrete form:

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)}, \qquad p(y) = \sum_{x_i \in \mathcal{X}} p(y \mid x_i)\, p(x_i)$$

Continuous form:

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)}, \qquad p(y) = \int_{\mathcal{X}} p(y \mid x')\, p(x')\, dx'$$

Normalization (evidence term):

Since the denominator $p(y)$ does not depend on x, it acts as a normalizer:

$$p(x \mid y) = \frac{1}{\int_{\mathcal{X}} p(y \mid x')\, p(x')\, dx'}\, p(y \mid x)\, p(x)$$

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Bayes Rule

- Plays a key role in many applications (e.g., robotics) and inference in general.
- If x is a quantity we would like to infer from y, the probability $p(x)$ is the **prior probability** and y is called the **data** (e.g., sensor measurement).
- The probability $p(x \mid y)$ is the **posterior probability**.
- The probability $p(y \mid x)$ is often called the **generative model**, since it describes how state variables X cause sensor measurements Y

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Expectation of a Random Variable

Definition: The **expectation** (or expected value) of a random variable X is the average value it takes under its probability distribution.

$$E[X] = \begin{cases} \sum_x x\, p(x), & \text{discrete} \\ \int x\, p(x)\, dx, & \text{continuous} \end{cases}$$

Key Properties:

- Linearity: $E[aX + bY] = a\, E[X] + b\, E[Y]$
- If X and Y are independent, $E[XY] = E[X]\, E[Y]$

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Entropy

Measures the **uncertainty** or **information content** of a random variable. The entropy of a probability distribution

$$\mathcal{H}_p(x) = E[-\log_2 p(x)]$$

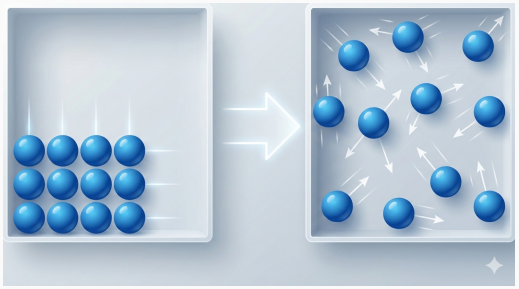
which resolves to

$$\mathcal{H}_p(x) = -\sum_x p(x)\ \log_2 p(x) \quad \text{(discrete case)}$$

$$\mathcal{H}_p(x) = -\int p(x)\ \log p(x)\, dx \quad \text{(continuous case)}$$

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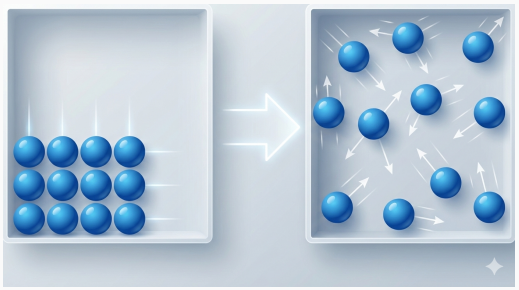
Entropy



- **Low entropy:** outcomes are predictable (distribution is concentrated) - little information gained from observation
- **High entropy:** outcomes are uncertain (distribution is spread out) - observations are informative

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Entropy



Complementary view:

- Entropy quantifies the expected number of **bits** required to uniquely identify the outcome of a random variable
- The expected number of binary (yes/no) questions needed to identify the outcome of a random variable.

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Entropy: Examples

How many bits would you need to encode each of these?

- Deterministic variable $\Pr(X = x) = 1$
- Fair coin $\Pr(X = h) = P(X = t) = \frac{1}{2}$
- Uniform over n outcomes $\Pr(X = x_i) = \frac{1}{n}$

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Entropy: Examples

How many bits would you need to encode each of these?

- Deterministic variable $\Pr(X = x) = 1 \Rightarrow \mathcal{H}(X) = 0$

  No uncertainty $\Rightarrow$ no bits needed

- Fair coin $\Pr(X = h) = P(X = t) = \frac{1}{2} \Rightarrow \mathcal{H}(X) = 1$

  One yes/no question $\Rightarrow$ one bit

- Uniform over n outcomes $\Pr(X = x_i) = \frac{1}{n} \Rightarrow \mathcal{H}(X) = \log n$

  $\log n$ bits distinguish 2^k possibilities

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Properties of Entropy

- $\mathcal{H}(X) \geq 0$
- Maximum entropy for uniform distributions
- Entropy increases with uncertainty
- Invariant to relabeling of outcomes

Interpretation: Entropy is the expected number of bits required to describe the outcome of X.

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Conditional Entropy

$$\mathcal{H}(Y \mid X) = \mathbb{E}_{x \sim p(x)} \big[\mathcal{H}(Y \mid X = x) \big]$$

- Measures remaining uncertainty in Y after observing X
- Conditioning never increases entropy:

$$\mathcal{H}(Y \mid X) \leq \mathcal{H}(Y)$$

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Joint Entropy and Chain Rule

- Joint entropy:

$$\mathcal{H}(X, Y) = -\sum_{x,y} \Pr(x, y) \log \Pr(x, y)$$

- Chain rule:

$$\mathcal{H}(X, Y) = \mathcal{H}(X) + \mathcal{H}(Y \mid X)$$

Total uncertainty decomposes into uncertainty before and after observing X.

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Why Entropy Matters

- **Learning:** uncertainty reduction
- **Inference:** confidence in predictions
- **Exploration (RL):** entropy regularization
- **Decision-making:** information gain

Entropy connects probability, information, and decision-making.

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Information Gain

Information gain measures the reduction in uncertainty about one variable after observing another.

$$I(Y; X) = \mathcal{H}(Y) - \mathcal{H}(Y \mid X)$$

- $\mathcal{H}(Y)$: uncertainty about Y before observing X
- $\mathcal{H}(Y \mid X)$: remaining uncertainty after observing X

Equivalent symmetric form: $I(Y; X) = \mathcal{H}(X) - \mathcal{H}(X \mid Y)$

Alternative expression: $I(Y; X) = \mathcal{H}(Y) + \mathcal{H}(X) - \mathcal{H}(X, Y)$

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Stochastic Dynamics

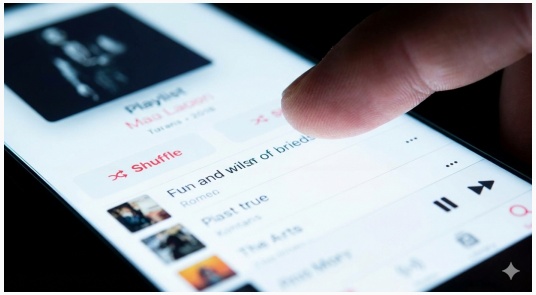
Stochasticity



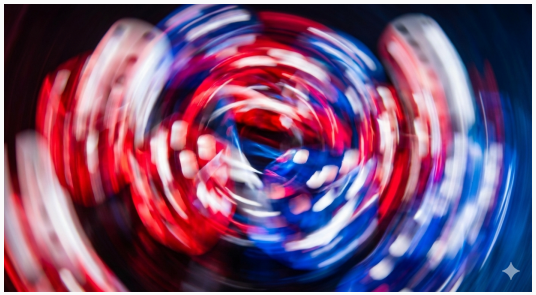
The outcome of a dice toss is governed by probability.

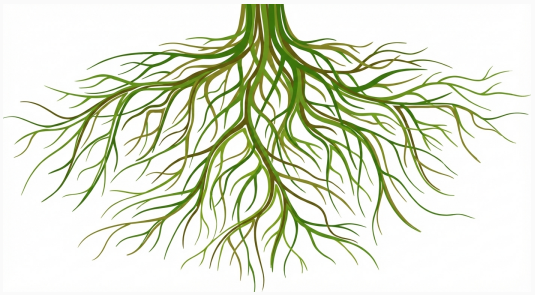


Growth is subject to a multitude of unpredictable factors like weather, soil quality, and pests.



Algorithms often employ stochastic processes to create variety (here- shuffling music)







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Stochastic Dynamics

- The environment evolves according to probability distributions
- Next state is sampled from a distribution (generative process).
- Captures uncertainty due to:
  - Noise
  - Unmodeled dynamics
  - Inherent randomness

Definition: Stochastic Process

A process associated with a probability function

$$\Pr[X_t = j \mid X_{t-1} = i, \ldots, X_0]$$

which describes how a system evolves under uncertainty and produces a sequence of random variables $X_0, \ldots, X_t$.

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Uncertainty

Suppose we observe the rainfall sequence:

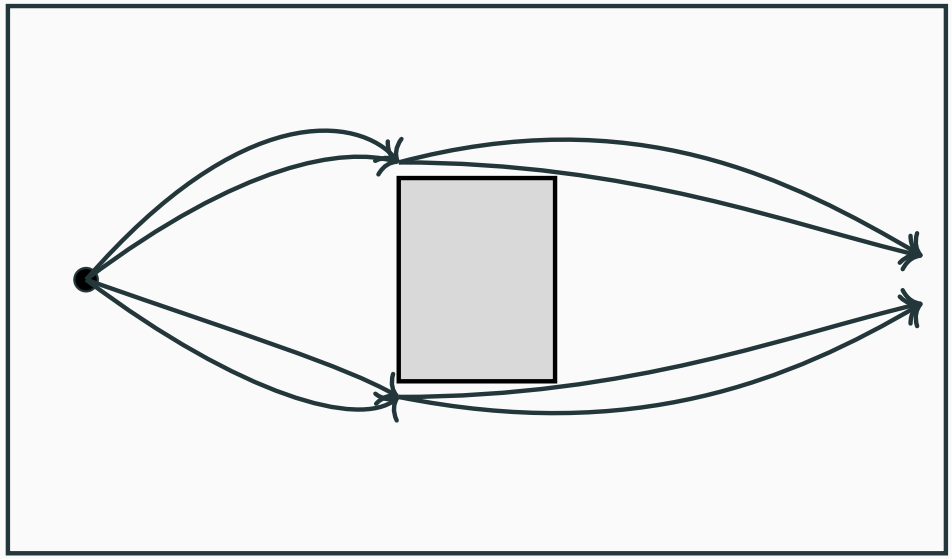
$$0,\ 0,\ 12,\ 0,\ 0,\ 15,\ 0,\ 0$$

Two explanations may exist:

1. The weather system is inherently stochastic.
2. There exists an unobserved atmospheric variable (e.g., a pressure system) influencing rainfall that is not included in the model.

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Uncertainty



People enter the room from a common entrance. To reach the exit, they must choose whether to pass the obstacle from the left or from the right.

Key Challenge: From observations alone, it may be difficult to determine whether the variability is due to **intrinsic randomness** or **missing knowledge about the system**.

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Aleatoric vs. Epistemic Uncertainty

Uncertainty captures why system evolution cannot be predicted.
Two (main) sources:

- Aleatoric Uncertainty (intrinsic randomness)

- Epistemic Uncertainty (lack of knowledge)

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Aleatoric vs. Epistemic Uncertainty

Uncertainty captures why system evolution cannot be predicted.
Two (main) sources:

- Aleatoric Uncertainty (intrinsic randomness)
  - Due to inherent stochasticity in the system
  - Cannot be eliminated, even with infinite data
  - Modeled directly by probability distributions

$$X_{t+1} \sim \Pr(\cdot \mid X_t)$$

- Epistemic Uncertainty (lack of knowledge)

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Aleatoric vs. Epistemic Uncertainty

Uncertainty captures why system evolution cannot be predicted. Two (main) sources:

- **Aleatoric Uncertainty (intrinsic randomness)**
  - Due to inherent stochasticity in the system
  - Cannot be eliminated, even with infinite data
  - Modeled directly by probability distributions

$$X_{t+1} \sim \Pr(\cdot \mid X_t)$$

- **Epistemic Uncertainty (lack of knowledge)**
  - Due to incomplete or imperfect knowledge of the model
  - Can be reduced with more data or better modeling
  - Common during learning or exploration

$$X_{t+1} \sim \Pr_{\theta}(\cdot \mid X_t) \qquad \theta \text{ not known}$$

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Aleatoric vs. Epistemic Uncertainty

Why is it important for us to differentiate between the two?

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Example: Aleatoric vs. Epistemic Uncertainty from Sequential Observations

Suppose there are two possible states: $\mathcal{S} \in \{s_1, s_2\}$ and two possible observations $\Omega \in \{o_1, o_2\}$

Initially, the state is not known, so both states are equally probable, i.e., $\Pr(S_0 = s_1) = \Pr(S_0 = s_2) = 0.5$

Assume we observe the sequence of observations:

$$O_{1:4} = (O_1 = o_1, O_2 = o_1, O_3 = o_1, O_4 = o_1)$$

Which is the more probable state after observing the sequence?

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Assume we observe the sequence of observations:

$$O_{1:4} = (O_1 = o_1, O_2 = o_1, O_3 = o_1, O_4 = o_1)$$

Which is the more probable state after observing the sequence?

And if we also know?

$\Pr(O_t = o_1 \mid S = s_1) = 0.8, \qquad \Pr(O_t = o_2 \mid S = s_1) = 0.2$

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Example: Aleatoric vs. Epistemic Uncertainty from Sequential Observations

Suppose there are two possible states: $\mathcal{S} \in \{s_1, s_2\}$ and two possible observations $\Omega \in \{o_1, o_2\}$

Initially, the state is not known, so both states are equally probable, i.e., $\Pr(S_0 = s_1) = \Pr(S_0 = s_2) = 0.5$

Assume we observe the sequence of observations:

$$O_{1:4} = (O_1 = o_1, O_2 = o_1, O_3 = o_1, O_4 = o_1)$$

Which is the more probable state after observing the sequence?

And if we also know?

$\Pr(O_t = o_1 \mid S = s_1) = 0.8, \qquad \Pr(O_t = o_2 \mid S = s_1) = 0.2$

And if we also know?

$\Pr(O_t = o_1 \mid S = s_2) = 0.1, \qquad \Pr(O_t = o_2 \mid S = s_2) = 0.9$

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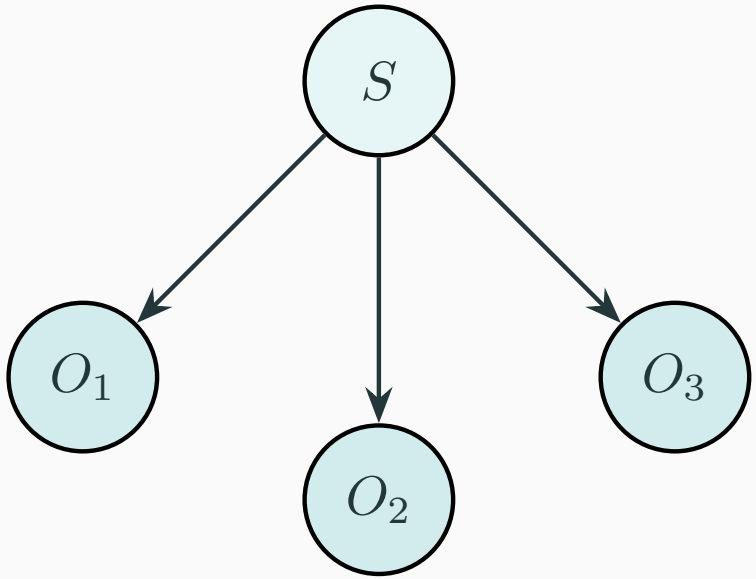
$\Pr(S_t = s_1 \mid S_{t-1} = s_1) = 0.01 \qquad \Pr(S_t = s_2 \mid S_{t-1} = s_2) = 0.01$

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Aleatoric vs. Epistemic Uncertainty

More data $\implies$ less epistemic uncertainty

More data $\not\Rightarrow$ less aleatoric uncertainty



Sequential observations help us infer **which source is active** (epistemic uncertainty), but they do not remove the **randomness of the observations themselves** (aleatoric uncertainty).

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Example - Newtonian dynamics with Stochastic Effects

Stochastic dynamics :

$$x_{t+1} = x_t + \Delta t \cdot \dot{x}_t + \eta_t, \quad \dot{x}_{t+1} = \dot{x}_t + \Delta t \cdot \ddot{x}_t + \epsilon_t$$

- ϵ_t is a random perturbation (e.g., zero-mean noise) on the velocity
- η_t represents noise on the position update.

Are ϵ_t, η_t aleatoric or epistemic sources of uncertainty?

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The Markov Property

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Definition: Markov Property

A stochastic process is *Markovian* if

$$\begin{aligned}\Pr(X_{t+1} = x' \mid X_t = x, X_{t-1}, \ldots, X_0) \\ = \Pr(X_{t+1} = x' \mid X_t = x)\end{aligned}$$

- The future is independent of the past, given the present (i.e., it depends only on the current state)
- Past (history) is summarized by the present

Markov Chains

Definition: Markov Chain

A stochastic process with:

- discrete time $t = 0, 1, 2, \ldots$
- discrete state space $\mathcal{X}$
- transition probabilities

$$\Pr(X_{t+1} = x' \mid X_t = x)$$

$p_{i,j}$ is the probability of transition from state i to state j

The probability function is a **transition matrix**

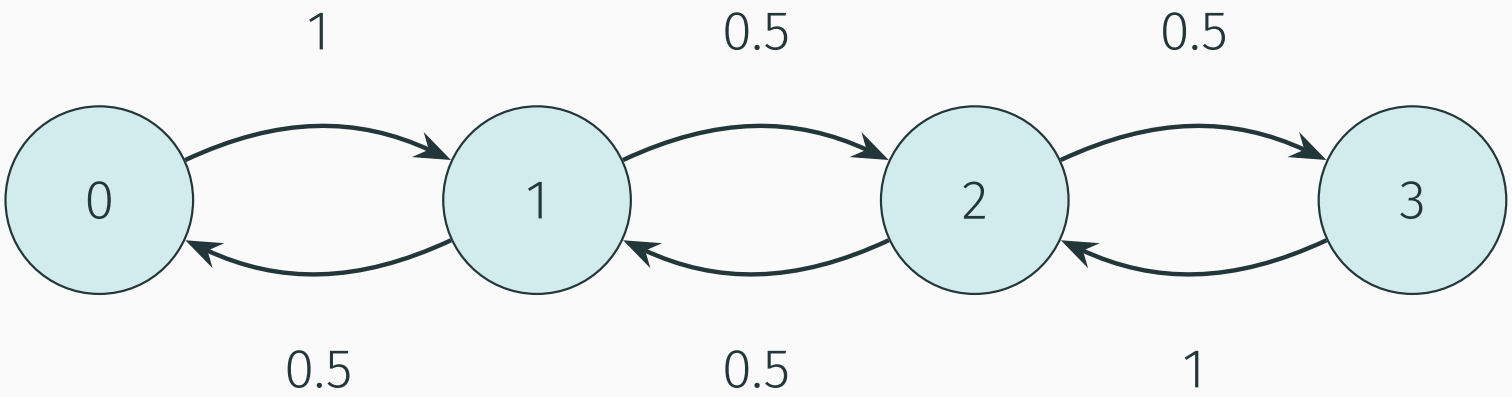
$$p_{i,j} \geq 0, \qquad \forall i \sum_j p_{i,j} = 1$$

We assume time homogeneity:

$$p_{i,j} = \Pr[X_t = j \mid X_{t-1} = i] = \Pr[X_1 = j \mid X_0 = i]$$

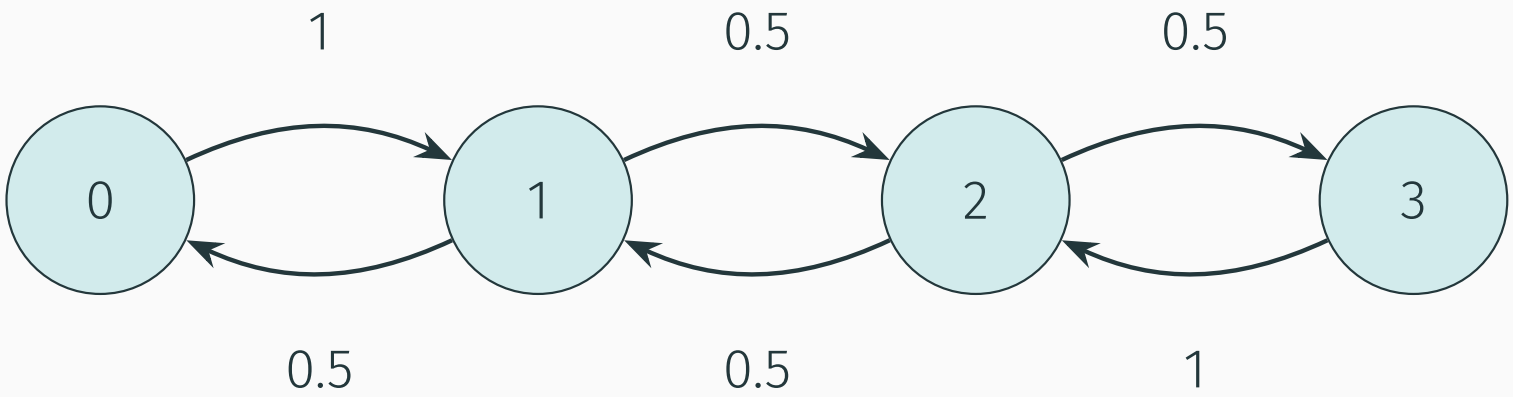
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Markov Chain Example



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Markov Chain Example



How do we compute the probability of a trajectory?

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Markov Chain: Probability of a Trajectory

- **Setup:**
  - Discrete-time Markov chain $\{X_t\}_{t \geq 0}$ on state space $\mathcal{X}$
  - Initial distribution

    $$\mu_0(i) = \Pr(X_0 = i)$$

  - Transition matrix P with entries

    $$p_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$$

- **Goal:** compute the probability of a trajectory

  $$\Pr(X_0 = i_0, X_1 = i_1, \ldots, X_t = i_t).$$

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Markov Chain: One-Step and m-Step Transitions

- Let P be the transition matrix of the Markov chain.
- **One-step transition probability:**

$$p_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$$

- **m-step transition probability:**

$$p_{ij}^{(m)} = \Pr(X_m = j \mid X_0 = i)$$

- **Matrix form:**

$$p_{ij}^{(m)} = (P^m)_{ij}$$

where P^m is the matrix product of P with itself m times.

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m-step Transition Probabilities

- Assume a time-homogeneous Markov chain with transition matrix P
- **m-step transition probability:**

$$\Pr(X_m = j \mid X_0 = i) = p_{ij}^{(m)}.$$

- **Expanded form:**

$$p_{ij}^{(m)} = \sum_{k_1,\ldots,k_{m-1}} p_{ik_1}\, p_{k_1k_2} \cdots p_{k_{m-1}j}$$

  - Sums over all possible intermediate state sequences
  - Captures the probability of reaching j from i in exactly m steps, where $k_1, \ldots, k_{m-1}$ denote intermediate states at times $1, \ldots, m-1$

- **Matrix form:**

$$P^{(m)} = P^m, \quad p_{ij}^{(m)} = (P^m)_{ij}$$

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Markov Chain: Probability of a Trajectory (Path Factorization)

Chain rule:

$\Pr(X_0 = i_0, \ldots, X_t = i_t) =$

$\Pr(X_0 = i_0) \prod_{k=1}^{t} \Pr(X_k = i_k \mid X_0 = i_0, \ldots, X_{k-1} = i_{k-1})$

Markov property:

$$\Pr(X_k = i_k \mid X_0 = i_0, \ldots, X_{k-1} = i_{k-1}) = \Pr(X_k = i_k \mid X_{k-1} = i_{k-1})$$

Result:

$$\Pr(X_0 = i_0, \ldots, X_t = i_t) = \mu_0(i_0) \prod_{k=1}^{t} p_{i_{k-1}, i_k}$$

This is the probability of starting at i_0 and then following the path $i_0 \to i_1 \to \cdots \to i_t$

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Example

Consider a two-state Markov chain, with transition probability P:

$$P = \begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix}$$

and initial distribution $p_0 = (0.5, 0.5)$

After one step, $p_1 = (0.3, 0.7)$

After two steps, $p_2 = (0.26, 0.74)$.

In the limit, $p_\infty = (0.25, 0.75)$ which is called the **steady-state distribution**, which represents the **long-run behavior of the system**, independent of the initial distribution. .

Why is the question of whether a steady state is reached important?

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Factorized Representations

Sequential Bayesian Updates

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Bayesian updating can be applied repeatedly as new data arrives

$$P(\text{Hypothesis} \mid \text{Data}_{1:t}) \propto P(\text{Data}_t \mid \text{Hypothesis}_t)\, P(\text{Hypothesis}_{t-1} \mid \text{Data}_{1:t-1})$$

Posterior is proportional to the likelihood times the prior:
Yesterday's posterior is today's prior.

Bayesian Networks

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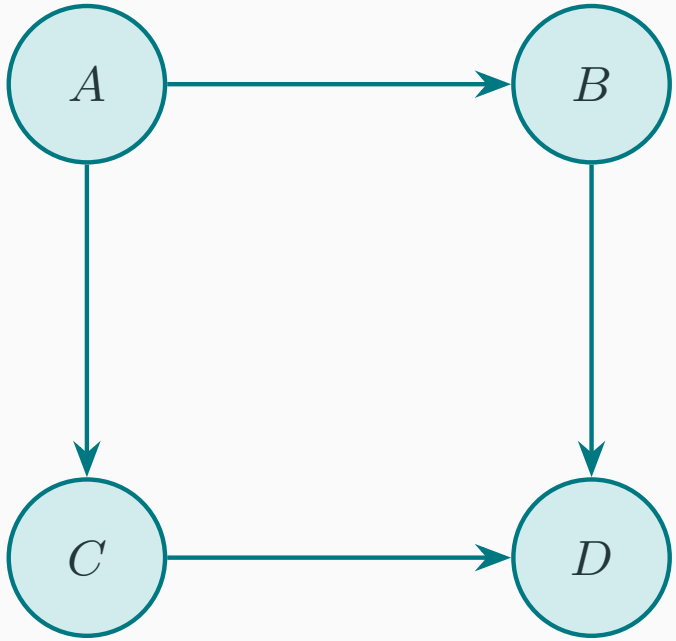
Definition: Bayesian Network (BN)

A directed acyclic graph (DAG) whose nodes are random variables and whose edges encode conditional dependencies.
It defines a joint distribution by factorization:

$$\Pr(X_1, \ldots, X_n) = \prod_{i=1}^{n} \Pr(X_i \mid \mathrm{Pa}(X_i)).$$

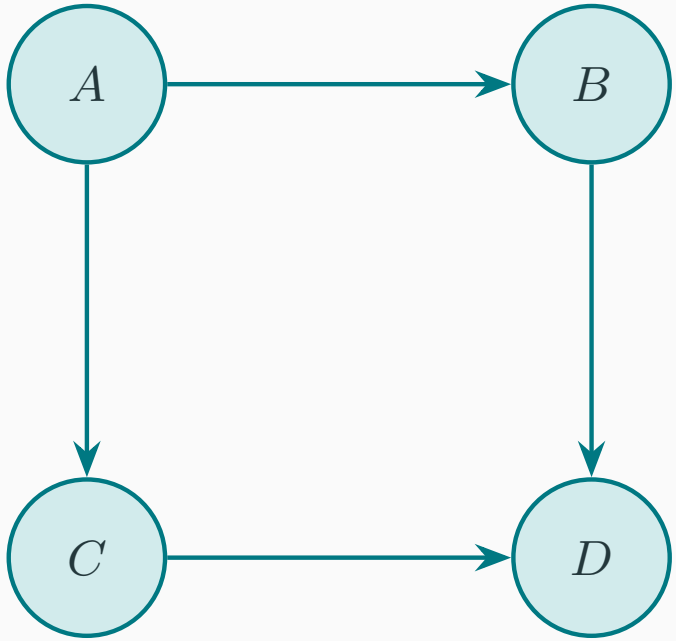
- $\mathrm{Pa}(X_i)$: parents of X_i in the DAG
- Useful for compact modeling and probabilistic inference

Bayesian Networks: Example



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Bayesian Networks: Example



$$\Pr(A, B, C, D) = \Pr(A)\ \Pr(B \mid A)\ \Pr(C \mid A)\ \Pr(D \mid B, C).$$

This is **graph factorization**, since we decompose the joint distribution into local conditional distributions defined by the graph structure.

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Bayesian Networks

- **Graph factorization:**

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P\big(X_i \mid \mathrm{Pa}(X_i)\big)$$

  The graph defines how the joint distribution decomposes into local conditional terms

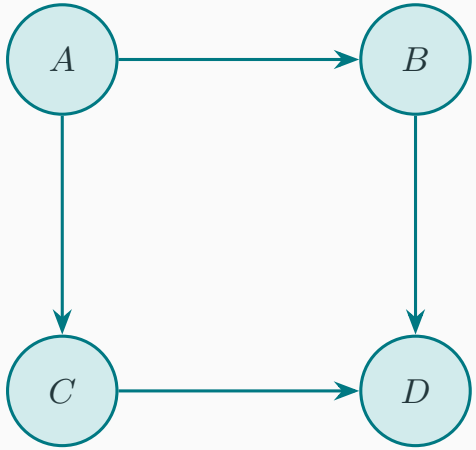
- **Edges encode direct dependence:** parents directly influence a node's distribution.
- **Local Markov property:**

$$X_i \perp\!\!\!\perp \text{NonDescendants}(X_i) \mid \mathrm{Pa}(X_i)$$

  X_i is conditionally independent of all its non-descendants, given its parents $\mathrm{Pa}(X_i)$.
- **Inference tasks:**
  - **Prediction:** compute $P(Y \mid X = x)$
  - **Diagnosis:** compute $P(\text{cause} \mid \text{effect})$
  - **Most likely explanation (MAP):** $\arg\max_z P(z \mid e)$ where z is the observation and e is the evidence.

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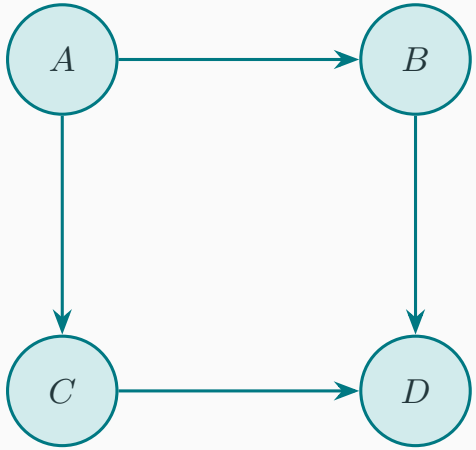
Bayesian Network Example: Semantic Meaning



- Let the variables represent:
  - A: whether a patient has a **viral infection**
  - B: whether the patient has a **fever**
  - C: whether the patient has a **cough**
  - D: whether the patient **visits a doctor**
- What does the graph encode?

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  - D: whether the patient **visits a doctor**
- What does the graph encode?
  - The infection can cause both fever and cough
  - Fever and cough jointly influence whether the patient visits a doctor
  - B and C are conditionally independent given A

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Dynamic Bayesian Networks (DBNs)

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Definition: Dynamic Bayesian Network (DBN)

A *Dynamic Bayesian Network* models a stochastic process evolving over discrete time. It is a Bayesian network repeated over time slices, with edges within a slice and across slices.

Let X_t be the set of variables at time t. A common (first-order) DBN specifies:

$$\Pr(X_0) \prod_{t=0}^{T-1} \Pr(X_{t+1} \mid X_t).$$

- First-order Markov: $\Pr(X_{t+1} \mid X_{0:t}) = \Pr(X_{t+1} \mid X_t)$
- Time-homogeneous (often): same Conditional Probability Distributions (CPDs) reused for all t

DBN Example: Same Semantics Over Time

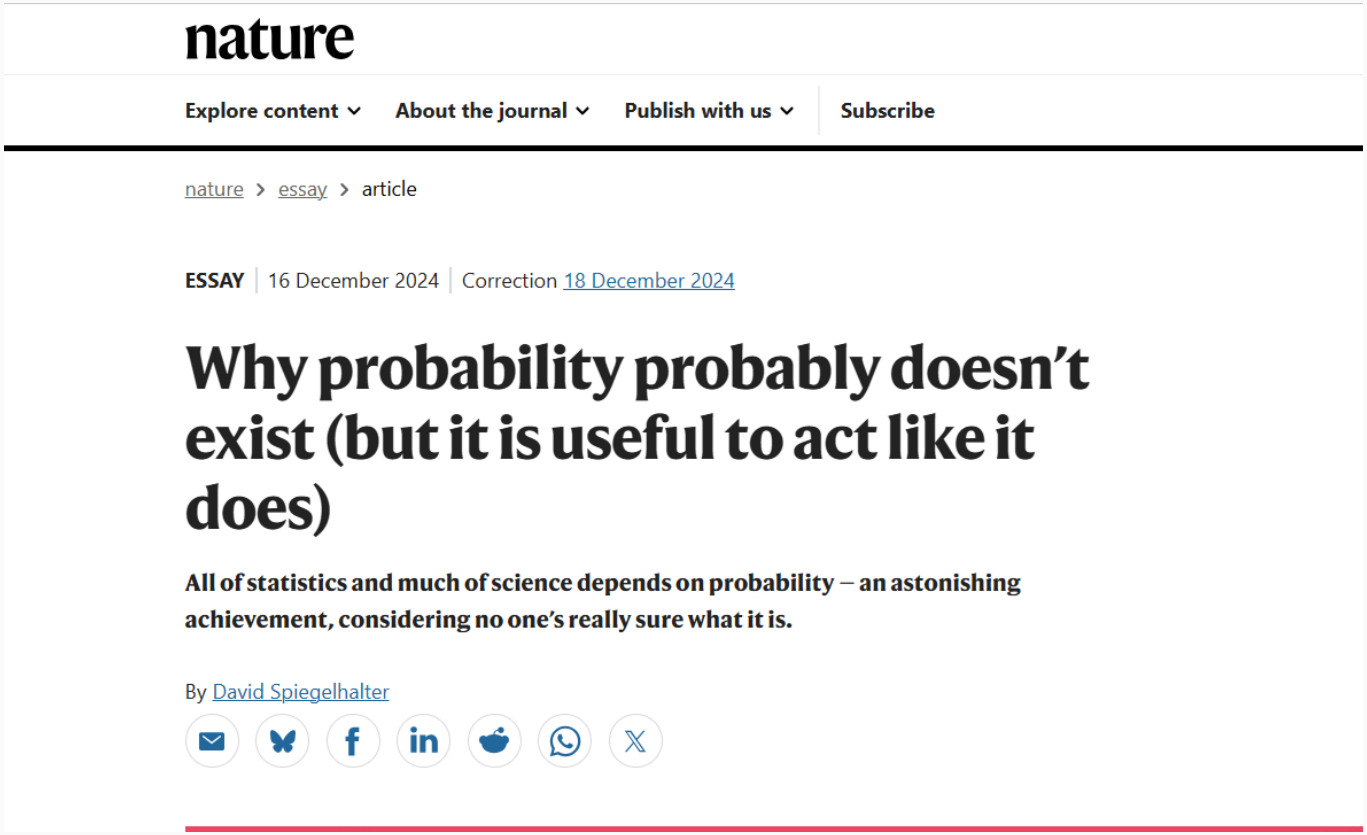
We reuse the same variables as before, now indexed by time t:

- A_t: whether A holds at time t
- B_t: whether the B holds at time t
- C_t: whether C holds at time t
- D_t: whether D holds at time t

The DBN assumptions:

$$\Pr(X_{t+1} \mid X_t) = \Pr(A_{t+1} \mid A_t)\ \Pr(B_{t+1} \mid A_{t+1})$$
$$\Pr(C_{t+1} \mid A_{t+1})\ \Pr(D_{t+1} \mid B_{t+1}, C_{t+1})$$

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ESSAY | 16 December 2024 | Correction 18 December 2024

Why probability probably doesn't exist (but it is useful to act like it does)

All of statistics and much of science depends on probability – an astonishing achievement, considering no one's really sure what it is.

By David Spiegelhalter

https://youtu.be/rcYM6XjRW-Q?si=1wlABo2mCIBdpxMz